Diego Méndez Carter

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Profile

Mechatronics Engineer (UNAM) and M.Eng. student in Robotics, Graphics and Computer Vision at Universidad de Zaragoza. Combines 2.5 years of industry experience qualifying electromechanical systems at Chamberlain Group with research in physics simulation, deformable object manipulation, and robot perception. Specialized in GPU-accelerated simulation routines, sensor data acquisition, and trajectory control for Sim-to-Real robotics pipelines.

Education

M.Eng. in Robotics, Graphics and Computer Vision — **Universidad de Zaragoza (UNIZAR)** Sep 2024 – Present

Zaragoza, Spain

- Coursework: Autonomous Robotics (ROS2), Advanced SLAM, Machine Learning, Computer Vision, Applications of Deep Learning, Parallel Computing.
- Thesis (In development): "Robotic Manipulation of Deformable Objects: An End-to-End Pipeline for Physics Simulation, Perception, and Control in Garment Folding"

B.S. in Mechatronics Engineering — **National Autonomous University of Mexico (UNAM)** Aug 2017 – Dec 2022

Mexico City, Mexico

- Thesis: Conceptual Design of a Beach Waste Collection System.
- Core Focus: Robotics Foundations, Systems Modeling, Rapid Prototyping, Embedded Systems.

Research Experience

Robotics Research Intern — **Universidad de Zaragoza (UNIZAR)** Feb 2026 – Jun 2026

Zaragoza, Spain

- Developed and validated deformable garment manipulation environments in the **Genesis physics engine**, evaluating PBD cloth solver accuracy against benchmark datasets (ClothesNet, FoldNet).
- Engineered GPU-accelerated simulation routines in **PyTorch/CUDA**, tuning substepping dynamics (Δt), contact stiffness, and friction parameters to resolve physical divergence.
- Implemented semantic perception pipelines and Cartesian trajectory tracking on **Franka Emika Panda** and **UR10e** manipulators to evaluate Sim-to-Real transfer feasibility.

Professional Experience

Sourcing Engineer (Electronic Components) — **Chamberlain Group** Jan 2023 – Jun 2025

Sonora, Mexico

- Performed circuit and gap analysis for component-level functional qualification across residential and commercial electromechanical drive systems (actuators, motors, passives).
- Executed root-cause failure analysis and bench testing alongside R&D Design, Manufacturing, and Quality teams under tight operational deadlines.
- Evaluated and validated novel electronic component technologies to support next-generation product designs, authoring formal technical qualification reports.
- Mentored and onboarded junior engineering interns through the Design Apprentice Program.

Selected Projects — M.Eng., UNIZAR

- **Autonomous Navigation Benchmarking:** Implemented A* and RRT global path planners as ROS2 Nav2 plugins; benchmarked execution latency and path optimality across diverse obstacle maps.
- **Computer Vision / SLAM:** Built a Structure-from-Motion pipeline and RGB-D odometry system (**Open3D**, **OpenCV**) incorporating sensor calibration, depth filtering, and point cloud registration.
- **CROFI Robotics Club (UNAM):** Designed and programmed competition robots (minisumo, RoboCup Junior) using Arduino, Raspberry Pi, OpenCV, ROS, and Fusion 360.

Skills

- **Robotics & Simulation:** Genesis World, ROS / ROS2 (Nav2), PBD cloth models, Open3D.
- **Programming & Compute:** Python, C++, PyTorch, CUDA, Linux, Git, Bash.
- **Perception & Vision:** OpenCV, RGB-D Odometry, Point Cloud Processing, SLAM, Sensor Calibration.
- **Hardware & Test:** Component testing, Circuit analysis, Oscilloscopes/DAQ, PCB review (Eagle), CAD (Fusion 360).
- **Languages:** Spanish (native), English (C1), French (A2).